

RIN RAY

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CRC Deformations-Geometry-Structure Postdoctoral Fellow, Universität Münster

ACADEMIC EMPLOYMENT

- 2023–Present **CRC Postdoctoral Fellow**; Universität Münster; Arithmetic and Homotopy Theory Working Group, with Thomas Nikolaus and Christopher Deninger
- chromatic homotopy theory · the Lubin–Tate action · moduli stacks of curves · arithmetic geometry
- 2015 **Visiting Researcher**; Santa Fe Institute of Complex Systems; Santa Fe, NM
- multiscale physics modelling · directed type systems
- 2013 **Computational Chemistry Researcher**; George Mason University, VA; advised by Dimitrios Papaconstantopoulos
- superconducting transition temperature · electronic-structure calculation · augmented plane wave method · composite materials
- 2013 **Automated Computational Semantics & Machine Learning Researcher**; George Mason University; Fairfax, VA
- probabilistic lexicon induction · natural language processing · decipherment
- 2012 **Human-Computer Interaction Researcher** (Autonomous Robotics); George Washington University; Washington D.C.
- proprioceptive-only control under graceful degradation · human–robot interaction
- 2012 **Autonomous Aerial Vehicle Researcher**; George Mason University
- simultaneous localization and mapping (SLAM) · motion planning · visual perception for unmanned aerial vehicles in novel environments
- 2011 **Research Assistant in Materials Science Lab**; Mary Baldwin College; Staunton, VA
- memristive devices · resistive switching

EDUCATION

- 2017–2023 **Northwestern**; PhD in Maths; advised by Paul Goerss
- A Geometric Model of Higher K-theories at Height $h=p^{k-1}(p-1)$ via Families of Ramified Curves*
- 2015–2017 **UChicago**; Divisional Masters Program in Maths; advised by Peter May and Kazuya Kato
- Calculating $\pi_*(tmf)$ at the prime 2*
 - Models of Formal Group Laws of Every Height*
- 2012–2013 **George Mason University**; B.S. in Computational Physics (graduated winter 2013, at 16); advised by Dimitrios Papaconstantopoulos
- Contextual Machine Learning through the Analysis and Chunking of Partially Translated Grade 2 Braille*
 - Predicting Superconductivity Transition Temperature via the Augmented Plane Wave Model*
- 2010–2012 **Mary Baldwin College**; Program for the Exceptionally Gifted: Mathematics, Computer Science, and Physics; advised by John Ong

PUBLICATIONS

- 2022 **Automorphisms of Abelian Varieties and Principal Polarizations**; joint with D. Lee; *Rendiconti del Circolo Matematico di Palermo*, Series 2, vol. 71, pp. 483–494; arXiv:1811.07007
- 2020 **Towards Directed Collapsibility**; joint with R. Belton, R. Brooks, S. Ebli, L. Fajstrup, B. T. Fasy, N. Sanderson, E. Vidaurre; *Advances in Mathematical Sciences*, vol. 21, pp. 255–271; arXiv:1902.01039

PREPRINTS

Sept 2026	On the Chromatic Splitting Conjecture at Coheight 1 ; joint with T. Barthel, L. Mann, A. Senger, T. Schlank, J. Weinstein, X. Zhou; on Jacquet–Langlands and homotopy theory
Sept 2025	Moduli Stacks of G-Curves in Homotopy Theory at $h = p-1$; arXiv:2509.23428
UPDATED SEPT 2026	
July 2025	Towards the $p = 3$ Kervaire Invariant Problem ; joint with E. Belmont; the E_2 -page for the homotopy fixed points spectral sequence computing $\pi_*(E_6^{hC_9})$; arXiv:2507.10157
July 2025	Modeling Group Actions on Stacks (Especially the Lubin–Tate Action) under construction ; arXiv:2507.00309
2018	A Global Crystalline Period Map ; joint with M. Neaton and A. Pieper; from the Arizona Winter School; arXiv:1911.08615

IN PROGRESS

Zeta Functions & Geometric Langlands	L-genera and localizations in K-theory ; joint with D. Berwick-Evans and N. Pacheco-Tallaj
Moduli Stacks & Geometric Langlands	Bounded modifications of G-curves and homotopy theory at $h = p^{k-1}(p-1)$
Zeta functions	Syntomic cohomology of ring spectra and a $T(h)$-local zeta function ; joint with G. Angelini-Knoll
Zeta functions	Toward categorifying the relationship of symplectic L-functions and Reidemeister torsion ; Riemann–Roch for Reidemeister torsion in K-theory and L-theory
Moduli stacks	The eigenvalues of Frobenius of Artin–Schreier–Witt curves are Gauss sums ; new families of Newton strata in the Torelli locus
Zeta functions	All Bernoulli numbers in homotopy theory are shifts on hiatus ; joint with A. Mejia and N. Riggenbach; connecting Kervaire–Milnor to Quillen–Lichtenbaum

FELLOWSHIPS

2022	Nominated for Presidential Fellowship ; Northwestern University
2017	Awarded NSF GRFP Fellowship (103K + 36K)
2014–2016	Awarded Thiel Fellowship (100K); mathematics and assistive technology; mentor Edward Frenkel

AWARDS

2022	Gelfand Prize , Northwestern University; for outstanding progress in one's own research and extraordinary contributions to the research goals of the graduate community
2013	Outstanding Rising Senior Award in Computational and Data Sciences, SPACS

SERVICE, TEACHING AND OUTREACH

Winter 2025	ORGANIZER Linearity in p-adic Lie Theory ; Homotopy and Arithmetic joint working group
Spring 2025	ORGANIZER K-Theoretic Poitou–Tate Duality ; Homotopy and Arithmetic joint working group
2025	ORGANIZER TEACHER “Mathe trifft Kunst” (Math Meets Art) ; activity designer and co-organizer; hands-on mathematics workshops for children, Universität Münster
Summer 2024	ORGANIZER Young Topologists Meeting
2024	PANELIST Queer in Math Day ; MPI Leipzig
Winter 2024	ORGANIZER Euler Characteristics and Zeta Functions ; homotopy working group

Winter 2023 **ORGANIZER** **K-Theory of the Integers**; Homotopy and Arithmetic joint working group

2022–2023 **FOUNDER** **Mental Health Math Buddy Program** **ongoing**; Northwestern. Pairs first- and second-year graduate students with older graduate students, meeting once a quarter for mental health check-ups.

2022 **FUNDING** Established the **Fund for Work–Life Activities for Graduate Students**; Northwestern University

Winter 2022 **ORGANIZER** **Stop Staring and Compute!**; interactive methods of computation in number theory and algebraic geometry

2021–2023 **MENTOR** **TGS Diversity Peer Mentorship Program**; Northwestern. Upper-year graduate students help first-year PhD students from diverse backgrounds in other subjects acclimate, providing regular check-ins, encouragement and support.

2021–2023 **MENTOR** **Causeway Postbaccalaureate Program**; Northwestern. Mentoring students from diverse backgrounds through a bridge programme preparing them for graduate school.

2021–2023 **MENTOR** **TEACHER** **Directed Reading Program**; Northwestern. Weekly one-on-one work with an undergraduate on a topic of their choice, ending in a talk to their peers.

2020–2023 **VOLUNTEER** **Evanston Animal Shelter**; fostered special-needs adult cats and raised neonatal kittens.

Winter 2020 **ORGANIZER** **Moduli Spaces**; alternating algebraic and analytic

Spring 2019 **ORGANIZER** **Applications of Topos Theory**; homotopy working seminar

Winter 2019 **ORGANIZER** **TamagaWHAT**; on the proof of the Tamagawa conjecture for function fields

Winter 2019 **ORGANIZER** **Formal Moduli Problems**; homotopy working seminar

Fall 2018 **ORGANIZER** **Geometry for Prime Addicts**; p -adic geometry toward Scholze's proof of the monodromy weight conjecture

Fall 2018 **ORGANIZER** **Factorization Homology**; homotopy working seminar

2016–2017 **TEACHER** **Math Circles**; weekly volunteering with children aged 6–12, South Side of Chicago

2015–2017 **TEACHER** **MENTOR** **Research Experience for Undergraduates**; University of Chicago. Summer programme with weekly lectures and interaction, guiding each undergraduate through a targeted individual research project.

2014 **FOUNDING CONTRIBUTOR** **Topos House** **ongoing**; a still-active house in South San Francisco for scientific and artistic collaboration, especially between mathematicians and biologists.

OTHER EMPLOYMENT

2014–2016 **Thiel Fellow** (Math and Assistive Tech Inventor); Thiel Foundation; Berkeley, CA
• medical and assistive technology · independent mathematics study under Edward Frenkel

2013–2014 **Mouse Vocalization Specialist & Audio Processing Engineer**; Vium; Belmont, CA
• computational bioacoustics · unsupervised classification of ultrasonic vocalizations (USVs) · quasi-real-time audio processing

2013 **Software Engineering Intern**; Cloudera; Palo Alto, CA
• Hadoop · Hive · large-scale data processing

SCIENTIFIC WORKS

2015 **Simplifying Multiscale Modeling**; Santa Fe Institute of Complex Systems

2014 **A New Female-Female Mouse Vocalization Discovered via Unlabeled Machine Learning**; Vium

2014 **On the Detection and Prevention of Aggression in Lab Mice via Quasi-Real Time Analysis**; Vium

2013 **Nonprovisional patent application**; Robotic Mobility Assistive Wheelchairs, 90776.0001USP1
• robotic mobility · pressure injury prevention

SELECTED INVITED TALKS

2026	Mod p Langlands and the Chromatic Splitting Conjecture ; Essen Number Theory Seminar
2024	Moduli Stacks of Curves in Homotopy Theory ; Johns Hopkins Topology Seminar
2021	Galois Theory of Spheres ; Front Range Number Theory
2020	Explicit Arithmetic of Jacobians ; ICERM Workshop on Arithmetic Geometry, Number Theory, and Computation
2019	Inverse Galois Theory in the Homotopy Groups of Spheres ; Princeton CCR Colloquium
2019	Galois Theory and Galois Representations in Homotopy Theory ; Higher School of Economics, Moscow
2019	An Introduction to the Classic Theory of p-Divisible Groups ; Oberwolfach Arbeitsgemeinschaft

CREATIVE WORKS

2026	Impaction , first play; staged in Los Angeles as part of <i>Nothing to See Here</i> , directed by Jacques Manjarrez
2025	Endomortis , comic; with Petra Flurin; crayolapunk surrealist comic about grief